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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 10/27/2025 |
| **PI Name:** | Mao, Xiao Wen M.D. | | |
| **Project Title:** | Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes<br>(2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | xmao@llu.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 909-558-8373 |
| **Organization Name:** | Loma Linda University | | |
| **PI Address 1:** | Radiation Medicine | | |
| **PI Address 2:** | 11175 Campus Street, Rm A1010, Chan Shun Pavilion | | |
| **PI Web Page:** | | | |
| **City:** | Loma Linda | **State:** | CA |
| **Zip Code:** | 92350-0001 | **Congressional District:** | 31 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship |
| **Start Date:** | 12/14/2017 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 09/30/2025 per NSSC information (Ed., 8/6/24)<br>NOTE: End date changed to 09/30/2024 per NSSC information (Ed., 11/9/23)<br>NOTE: End date changed to 09/30/2023 per C. Ribeiro/JSC and NSSC information (Ed., 4/7/22)<br>NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21)<br>NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21)<br>NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20)<br>NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sun, Shu-Wei Ph.D. ( Loma Linda University ) | | |

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| **Grant/Contract No.:** | 80NSSC18K0310 |
| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent.<br>The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated.<br><br>Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures. |
| **Task Progress:** | This grant was completed by 9/30/2025. We have accomplished what we proposed in the grant.<br>In our MHU-8 flight study, we found that the optic nerve (ON) and retina (RTN) exhibited mostly distinct transcriptomic responses. The cellular components and structure of the ON and RTN are notably different in terms of cell types, astrocytic features, vascular density, and metabolic activity, as well as permeability and blood barrier properties. The largely unique gene expression profiles observed in this study in the ON and RTN point to a complex response to space-specific environmental insults. Further studies are needed to investigate the underlying mechanisms linking the observed transcriptomic changes to the development of pathologic neuro-ophthalmic abnormalities.<br><br>In the present radiation study, we found that low-dose GCRsim radiation induced significant changes in the expression of several proteins related to blood-retinal barrier (BRB) integrity, including APQ-4, PECAM-1, and ZO-1 in the retina. AQP-4, a water channel protein, has an important role in maintaining the integrity of the BRB by forming a functional complex at the astrocytic endfeet. The upregulation of AQP-4 in irradiated retinal ganglion cell layer (GCL) and inner nuclear layer (INL) indicates the potential disruption of barrier integrity and even possible edema formation. The degradation of TJ-associated proteins in the retina following injury has been shown to lead to BRB breakdown. In the current study, the decrease in ZO-1 protein expression in the retina, together with the increase in water channel protein AQP-4 and adhesion molecules PECAM-1, in conjunction with increased apoptosis, suggests the BRB disturbance and prolonged tissue remodeling due to radiation exposure.<br><br>Sex is one of the confounding factors that impact biological response to radiation exposure. In our mouse study, we found that some of the protein expression levels for BRB integrity following GCR exposure in the male mice were different from those of the female mice. Our data reveal a complex response to radiation exposure between female and male mice. It is possible that our findings may not lead to the conclusion that there is a direct correlation between sex and radiation-induced ocular damage, or if sex has a role as a risk factor. However, our study indicates some biomarkers related to BRB integrity are differentially regulated in the male mice versus females, which supports a potential role for gender-specific effects of radiation exposure. |
| **Bibliography Type:** | Description: (Last Updated: 07/10/2026) |
| **Articles in Peer-reviewed Journals** | Roy U, Hadad R, Rodriguez AA, Saju A, Roy D, Gil M, Keane RW, Scott RT, Mao XW, de Rivero Vaccari JP. "Effects of space flight on inflammasome activation in the brain of mice." Cells. 2025 Mar 12;14(6):417. https://doi.org/10.3390/cells14060417 ; PubMed PMID: 40136666; PubMed Central PMCID: PMC11941215 , Mar-2025 |
| **Articles in Peer-reviewed Journals** | Kremsky I, Pergerson R, Justinen S, Stanbouly S, Willey J, Fuller CA, Takahashi S, Vitaterna MH, Bouxsein M, Mao X. "Artificial gravity attenuates the transcriptomic response to spaceflight in the optic nerve and retina." Int J Mol Sci. 2024 Nov 9;25(22):12041. https://doi.org/10.3390/ijms252212041 ; PubMed PMID: 39596110; PubMed Central PMCID: PMC11593819 , Nov-2024 |
| **Articles in Peer-reviewed Journals** | Patel CM, Wiele SV, Kim L, Payne E, Bruno-Garcia M, Devorak A, Kaganov DE, Lau A, Guthold M, Delp MD, Crapo J, Mao XW, Willey JS. "Treatment with a superoxide dismutase mimetic for joint preservation during 35 and 75 days in orbit aboard the international space station, and after 120 days recovery on Earth." Life Sci Space Res. 2025 Feb;44:67-78. https://doi.org/10.1016/j.lssr.2024.10.009 ; PMID: 39864914 , Nov-2025 |